

COMP 8003

Lab 5

Design

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What it does

Goal

Demonstrates the weaponization phase of the cyber kill chain by designing and executing a benign activity that creates a hidden directory, and then copies */etc/passwd* into the directory.

Success condition (observable outcome)

The hidden directory, *hidden_dir*, is created in the user's home directory and */etc/passwd* is copied into this directory. The operations required to copy */etc/passwd* is written to a log file with *inotifywait*.

Invariant (one checkable rule)

The script always operates relative to the user's home directory as resolved from `~`.

How I will test it

Run script as a regular user

Script is run from the command line.

Steps:

1. Set the script to executable.
2. Monitor */etc/passwd* with *inotifywait*, then redirect and append to a log.
3. Run the script.
4. Terminate *inotifywait*.
5. Analyze *inotifywait* log.

Expected result:

- *.hidden_dir* is created in the user's home directory and */etc/passwd* is copied into this directory
- The operations required to copy */etc/passwd* is written to a log file with *inotifywait*.

What it proves:

- Script creates a hidden directory, which is then used to hold a copy of */etc/passwd*.
- Keeps */etc/passwd* unchanged.
- Terminates cleanly.

Run script with existing `.hidden_dir`

The script is run with an existing target hidden directory, `.hidden_dir`

Steps:

1. Run the script as a regular user.
2. Delete the `/etc/passwd` copy within `.hidden_dir`.
3. Run the script as the same user again.

Expected result:

- Running the script will produce an error indicating an existing `.hidden_dir`.
- `cp /etc/passwd ~/.hidden_dir` still executes and creates a new `passwd` file.
- `inotifywait` log will display access and close operations caused by the copy `/etc/passwd` line within the script.

What it proves:

- The script will fail to create another `.hidden_dir`.
- `cp /etc/passwd ~/.hidden_dir` will still be executed as normal following the hidden directory creation command.

Run script with existing `/etc/passwd` copy within `.hidden_dir`

The script is run with an existing `/etc/passwd` copy within `.hidden_dir`

Steps:

1. Run the script as a regular user.
2. Run the script as the same user again.

Expected result:

- Running the script will produce an error indicating an existing `.hidden_dir`.
- `cp /etc/passwd ~/.hidden_dir` still executes and overwrites the existing `passwd` file.
- `inotifywait` log will display access and close operations caused by the copy `/etc/passwd` line within the script.

What it proves:

- The script will fail to create another `.hidden_dir`.
- `cp /etc/passwd ~/.hidden_dir` will still be executed as normal following the hidden directory creation command.
- `cp` overwrites existing files.

Run script with insufficient write permissions to the home directory

The script is run with insufficient write permissions relative to the home directory.

Steps:

1. Set the home directory permissions to 500 (readable and executable by the user).
2. Run the script as a regular user.

Expected result:

- *inotifywait* fails depending on output log location.
- *inotifywait* shows open and close operations to */etc/passwd*.
- *mkdir ~/.hidden_dir* fails because of insufficient permissions.
- *cp /etc/passwd ~/.hidden_dir* fails because the copied file cannot be created in *.hidden_dir*

What it proves:

- Script will fail to create *.hidden_dir* and the copy of */etc/password* due to lack of write permissions.
- Script still accesses */etc/passwd* in the line *cp /etc/passwd ~/.hidden_dir*, as displayed in the *inotifywait* output.

Run script with \$HOME unset

The script is run with \$HOME environment variable unset.

Steps:

1. Run the script with \$HOME unset, *env -u HOME ./weaponized_action.sh*

Expected result:

- *inotifywait* shows open and close operations to */etc/passwd* during the copy attempt.
- *mkdir ~/.hidden_dir* fails because ~ cannot be resolved.
- *cp /etc/passwd ~/.hidden_dir* fails because the target directory does not exist.

What it proves:

- Script will fail to create *.hidden_dir* and the copy of */etc/passwd* because ~ cannot be resolved.
- Script still accesses */etc/passwd* in the line *cp /etc/passwd ~/.hidden_dir*, as displayed in the *inotifywait* output.

FSM

State meanings

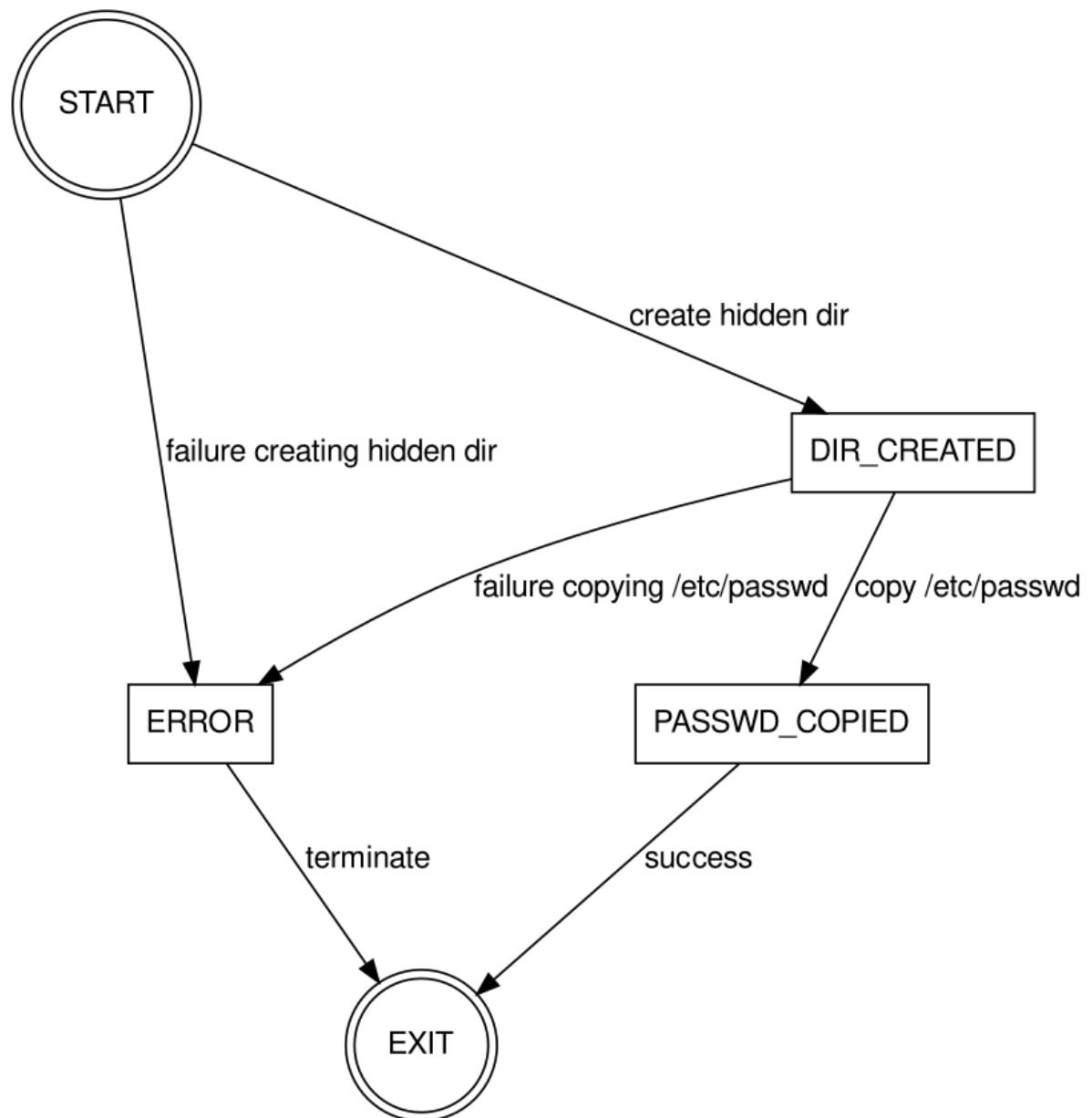
State	Meaning
START	The program has begun execution, but no work has been performed.

DIR_CREATED	Hidden directory has been successfully created.
PASSWD_COPIED	/etc/passwd successfully copied into the hidden directory.
ERROR	Failure occurred.
EXIT	The program has terminated.

Transition table

State	Call (function)	Fail state	Success state
START	mkdir ~/.hidden_dir	ERROR	DIR_CREATED
DIR_CREATED	cp /etc/passwd	ERROR	PASSWD_COPIED
PASSWD_COPIED			EXIT
ERROR	cleanup		EXIT

FSM diagram



Functions

Name	Inputs	Outputs	Responsibility
N/A			